DPUSteeringFFB

This DPU calculates a simple steering force model, which can be parameterized depending on the driving speed.

The steering force model consists of the three components

- Friction [*Friction_T*]
- Spring stiffness [*Stiffness_T*]
- Damping [*Damping_T*]
- and the additional torque provided as an optional input [*Torque*]

The resulting steering torque can be scaled by means of a parameter (Gain).

$$Torque_{Out} = (Torque + Friction_T + Stiffness_T + Damping_T) \cdot Gain$$

The calculation of the individual portions can be found in the functional description.

The dynamic parameters of the steering torque generation are partly stated without dimensions, as the actual dimension depends on the implementation in the actuator. Instead, the symbol used in the formula below the table is stated in square brackets.

1. Inputs:

SteeringWheel	Steering wheel angle in degrees or radians (the parameter *SteeringInRad* can be used to select which unit is used). Internal calculation is performed in radians [*SW* in rad]
SteeringWheelV	Optional steering wheel angular velocity in degrees/s or rad/s (see below). If this input is not connected, then the steering wheel angular velocity is generated from the filtered values of the steering wheel angle. Internal calculation is performed in rad/s [*SW_V* in rad/s].
SteeringInRad	This parameter can be used to determine whether the steering wheel angle (and steering wheel angular velocity) inputs are interpreted in rad (=*true*) or degrees (=*false*).
V_kmh	Vehicle velocity in [km/h]
Gain	Overall gain of the steering torque model.
Torque	Additional torque added to the calculated steering force model [*Torque*].

2. Dynamic parameters:

Friction	Friction parameter [*Fr*].
FrictionGain	Friction gain factor [*Fr_Gain*]

FrictionDelta	Static friction offset [*Fr_Delta*]
FrictionP	Mixing ratio of the dynamic portion to the static offset of the friction [*Fr_P* in the range 0…1]
Stiffness	Spring stiffness depending on the steering wheel angle [*St*]
StiffnessGain	Gain factor of spring stiffness [*St_Gain*]
StiffnessDelta	Static portion of spring stiffness [*St_Delta*]
StiffnessLin	Steering wheel angle from which the spring stiffness remains constant [*St_Lin* in rad]
Damping	Damping components depending on the steering wheel angular velocity [*Da*]
DampingGain	Gain factor of damping [*Da_Gain*]
DampingP	Mixing ratio of the linear portion of damping to the square portion of damping [*Da_P* in the range 0…1]

3. Static parameters:

vSLUT,vFLUT,vDLUT	Velocity-dependent Lookup tables for the *Stiffness* (vSLUT), Friction (vFLUT) and *Damping* (vDLUT) parameters. The interpretation of the parameters is determined by the settings in the Mode parameter. Assignment takes place in a tuple with pairs of velocity and parameter values (see example in the functional description).
Mode	=0: Normal mode, no velocity-dependent Lookup tables are used. =1: The parameters Friction, Stiffness and Damping are **multiplied** by the velocity-dependent Lookup table. =2: The parameters Friction, Stiffness and Damping are **replaced** by the values stated in the velocity-dependent Lookup table.
SteeringWheelFilterLength	Number of values considered for the calculation of the mean value of the internal steering wheel angle.
SteeringWheelVFilterLength	Number of values applied for averaging the calculation of the steering wheel angular velocity.
PositionMax	End stop position in degrees.
EndStopPosDelta	Relative transition point in degrees at which the impact force increases.
EndStopPosTorque	Restoring force for the end stop.
OutMax	Limitation of the steering torque. The output steering torque is limited to –*OutMax* … +*OutMax*.

4. Outputs:

TorqueOut	Resulting steering torque (calculation see functional description) [*Torque_Out*]
SteeringWheelOut	Filtered steering wheel angle in [rad]
SteeringWheelVOut	Filtered or calculated steering wheel angular velocity in [rad/s]
FrictionOut	Currently applied value for the parameter *Friction*
StiffnessOut	Currently applied value for the parameter *Stiffness*
DampingOut	Currently applied value for the parameter *Damping*. These outputs can be used to control the possibly velocity dependent Lookup tables.

FrictionTOut	Calculated portions of the friction,
StiffnessTOut	spring stiffness and
DampingTOut	damping components in the output steering torque.

5. Function:

The steering force model works internally with a steering wheel angle in rad and the steering wheel angular velocity in rad/s. The internally used steering wheel angle is initially averaged from the previous values (determined by *SteeringWheelFilterLength*). If the input *SteeringWheelV* is not connected, then the numerical derivative of the internal steering wheel angle is used as the internal steering wheel angular velocity and also filtered (determined by *SteeringWheelVFilterLength*).

The individual components of the steering torque feedback are calculated from

$$Friction_T = \left((|SW_V| \cdot Fr \cdot Fr_P) + (Fr_{Delta} \cdot Fr \cdot (1 - Fr_P))\right) \cdot Fr_{Gain}$$

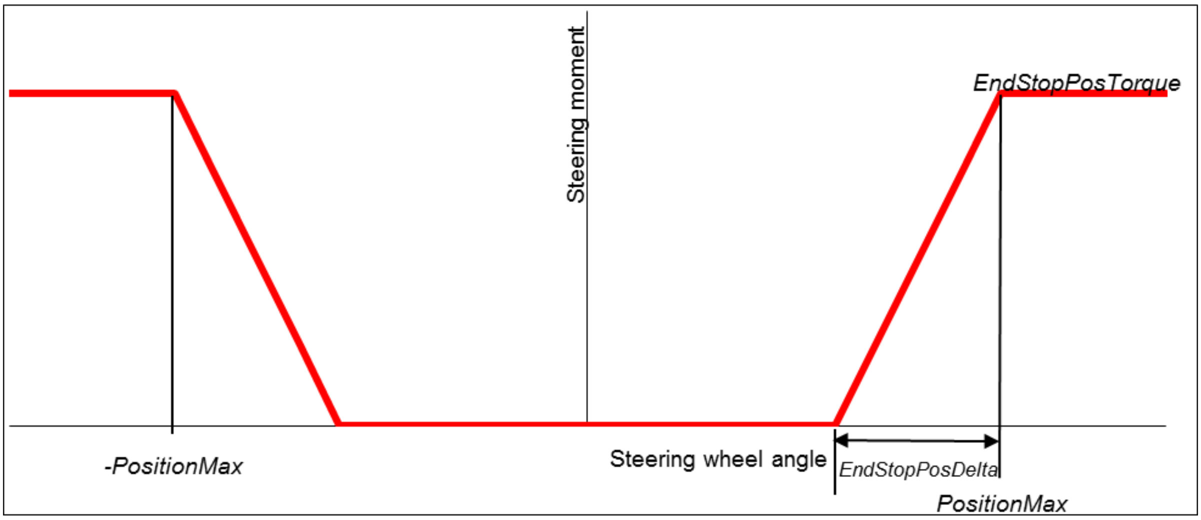
$$Stiffness_T = \begin{cases} (-SW_{Lin} \cdot St - St_{Delta}) \cdot St_{Gain} & | \quad SW < -SW_{Lin} \\ (SW \cdot St - St_{Delta}) \cdot St_{Gain} & | \quad SW < 0 \\ (SW \cdot St + St_{Delta}) \cdot St_{Gain} & | \quad SW > 0 \\ (SW_{Lin} \cdot St + St_{Delta}) \cdot St_{Gain} & | \quad SW > SW_{Lin} \end{cases}$$

$$Damping_T = (|SW_V| \cdot Da \cdot Da_P + SW_V^2 \cdot Da \cdot (1 - Da_P)) \cdot Da_{Gain}$$

The resulting steering torque

$$Torque_{Out} = (Torque + Friction_T + Stiffness_T + Damping_T) \cdot Gain$$

is superimposed by the following curve, in order to simulate an end stop for the steering wheel angles beyond the area determined by *PositionMax*:

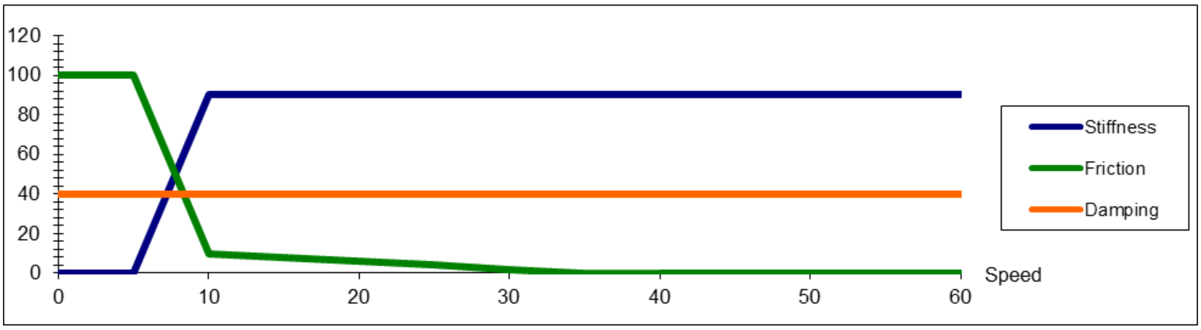


The parameters *Friction*, *Stiffness* and *Damping* can also be determined using velocity-dependent Lookup tables.

A drilling torque (= increased friction) in the lower velocity range as well as a continuous increase of the restoring force in the upper velocity range can, for example, be achieved with the following parameterization:

```
DPUSteeringFFB ~SteeringFFB
{
      Index = 5;
      Computer = {MOCKUP};
      Gain = 500;
      Mode = 2;
      vSLUT = (  0.0,  0.0,
                5.0,  0.0,
                10.0, 90.0
            );
      vFLUT = ( 0.0, 100.0, # 0…5 km/h:   100.0
                5.0, 100.0, # up to 5 km/h: 100.0
               10.0,  10.0,
                  # drop sharply in range 5 km/h to 10 km/h
               35.0,   0.0
                  # allow to drop from 10 to 0 at higher speeds
            );
      vDLUT = ( 0.0, 40.0,
              100.0, 40.0
      );
};
Connections =
{
      VDyn.v_kmh -> ~SteeringFFB.v_kmh,
      Mockup.SteeringWheel -> ~SteeringFFB.SteeringWheel,
      ~SteeringFFB.TorqueOut -> ~Lenkmotor.KraftIn
};
```

In the example above, the parameter curves depicted in the following graph are realized. The parameters in the tuple assignment each represent pairs of velocity and parameter values for a node and must be stated in ascending velocity. Linear interpolation is used between the nodes determined in this way. The last assumed value remains if the speed exceeds the defined range.



As *Mode = 2* is used, the *Friction* parameters are defined via the table *vFLUT*, *Stiffness* via the table *vSLUT* and the *Damping* parameter via *vDLUT*. In this mode, the inputs *Friction*, *Stiffness* and *Damping* of the DPU are ignored.

For example, a value of 100.0 is stated for *vFLUT* for friction and for the velocity value 0.0 with the first pair of values in the tuple assignment. This value is maintained up to a velocity of 5.0.

Between 5.0 and 10.0, the friction is faded over from 100.0 to 10.0 in a linear manner and then completely faded out to 0 up to a velocity of 35.0. The last stated value of 0.0 for friction applies for velocities exceeding 35.0.

The spring stiffness is faded in to the value 10.0 in a range of 5.0 … 10.0 via the table vSLUT. Damping is set to a constant value of 40.0 via vDLUT.